



The Role Of Cooperative Learning Strategy In Developing Linguistic Intelligence Among Primary School Pupils From Teachers' Perspective

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Abstract ;

This field study aimed to investigate the role of cooperative learning strategy in developing linguistic intelligence among primary school students from teachers' perspective. Using a descriptive-analytical design, the study surveyed 114 primary school teachers in Djelfa Province during the 2024/2025 academic year. A validated 20-item electronic questionnaire (Cronbach's alpha = 0.695) was employed. Results revealed statistically significant gender differences in the use of cooperative learning strategies favoring male teachers, and confirmed a positive correlation between the degree of strategy implementation and students' linguistic intelligence levels. The study concluded that cooperative learning effectively enhances oral and written communication skills and fosters critical and creative thinking. Recommendations include professional development for teachers in diversifying cooperative strategies, integrating them into teacher training curricula, and developing direct assessment tools to measure their impact on students' linguistic intelligence.

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1. Introduction

Early childhood education constitutes one of the most fundamental pillars upon which students' cognitive and skill development is built. Effective instructional strategies contribute to enhancing learners' abilities and fostering their multiple intelligences, particularly linguistic intelligence, which serves as an essential tool for communication and critical thinking. In this context, cooperative learning emerges as a foremost modern pedagogical strategy aimed at actively engaging students in the educational process through collaborative work, experience sharing, and problem solving within a stimulating, participatory environment.

Primary school teachers play a central role in implementing this strategy, as they are responsible for selecting appropriate activities, assigning roles among students, and accommodating individual differences, all of which effectively contribute to the development of their linguistic skills. This responsibility assumes special significance given the increasing emphasis on cultivating students' oral and written communication skills, which bolster self-confidence, stimulate critical thinking, and enhance the capacity for accurate expression.

This field study aims to examine the role of the cooperative learning strategy in developing linguistic intelligence among primary school students from their teachers' perspective, and to assess the impact of this strategy on enhancing this form of intelligence through a purpose-designed questionnaire. It further seeks to analyze the extent to which variables such as teachers' educational level, professional experience, and gender influence the implementation of these strategies, thereby providing a knowledge base that may

contribute to improving pedagogical practices and enriching educational quality.

Problem Statement

Despite the growing interest in active teaching strategies, including cooperative learning, there remains variability in how teachers employ them, divergent opinions regarding their effectiveness and importance, and inconsistent evidence concerning their contribution to the development of linguistic intelligence, especially at the primary level.

Research Questions

- Are there statistically significant differences in teachers' use of the cooperative learning strategy attributable to gender?

- To what extent does the cooperative learning strategy influence the development of linguistic intelligence among primary school students, from their teachers' perspective?

Objectives

To identify the current use of the cooperative learning strategy by primary school teachers.

To measure the impact of the cooperative learning strategy on students' linguistic intelligence from the teachers' perspective.

To detect differences in the use of these strategies according to demographic variables.

Significance of the Study

Theory: To enrich the literature on cooperative learning and linguistic intelligence.

Practical: To provide field data for decision-makers and educational inspectors to improve pedagogical training.

Pedagogical: To propose an effective framework for activating strategies that develop linguistic skills among primary students.

Hypotheses

There are statistically significant differences in primary teachers' use of

cooperative learning strategies attributable to gender.

There is a positive correlational relationship between the degree of use of cooperative learning strategies and the level of linguistic intelligence among primary school students from their teachers' perspective.

Methodology

Research Design: Descriptive-analytical.

Population: All primary school teachers in the schools of Djelfa Province (2024/2025).

Sample: 114 teachers from five primary schools.

Study Instrument and Statistical Analysis

The study instrument was an electronic questionnaire comprising 20 items, whose validity and reliability were confirmed (Cronbach's $\alpha = 0.695$).

Statistical analysis included frequencies, percentages, means, standard deviations, independent-samples t-tests, and Pearson's correlation coefficient, all conducted using SPSS version 23.

Axis I: Nature of Cooperative Learning and Linguistic Intelligence

1. Concept of Cooperative Learning

Cooperative learning is a prominent contemporary pedagogical strategy in the field of education. It is based on engaging learners in small, heterogeneous groups whose members share roles and responsibilities to achieve common objectives. Empirical studies have demonstrated its effectiveness in developing learners' thinking and communication skills, thereby fostering positive interaction among peers. One of the dimensions influenced by cooperative learning is linguistic intelligence—one of the facets in Gardner's multiple intelligences theory—since it relates to an individual's ability to use language for expression, comprehension, and communication. Hence, it is important to

study the essence of cooperative learning and its relationship to the development of learners' linguistic intelligence in order to elucidate the theoretical frameworks underpinning both constructs.

Definitions from the Literature

Johnson & Johnson (1999) define cooperative learning as an educational environment in which learners work in small groups sharing responsibility to achieve common goals within a context of positive interaction and interdependence.

Slavin (2005) describes it as a teaching strategy that divides students into small, ability-diverse groups who collaborate to complete a specific academic task, with each individual's success tied to group success.

Abu Jaber (2009) characterizes cooperative learning as a modern instructional method involving students in collaborative workgroups, where each member bears responsibility for their role, contributing to the group's collective educational goal.

Zaitouni (2014) views it as a learning process that provides opportunities for interaction and communication among learners to construct knowledge and develop cognitive and social skills in an organized group context.

2. Linguistic Intelligence

Linguistic intelligence is one of the principal types of intelligence outlined in Gardner's theory of multiple intelligences. It pertains to an individual's capacity to use language effectively for expressing ideas and understanding others. This form of intelligence manifests in speaking, writing, reading, and listening skills, enabling effective communication and social influence (Gardner, 1983, p. 67). It is further defined as the ability to employ words and linguistic symbols in constructing meaning and facilitating effective communication, making it fundamental to success in education, literature, media, and professions requiring

advanced language use (Shaaban, 2006, p. 112). From an educational psychology perspective, linguistic intelligence is evidenced by learners' abilities to formulate sentences, comprehend and analyze texts, as well as to persuade and convey ideas clearly, reflecting an integration of cognitive capabilities and communicative skills (Mar'i & Al-Heila, 2010, p. 95).

3. Importance of Linguistic Intelligence

Linguistic intelligence serves as the primary tool for interpersonal communication and knowledge acquisition and transmission. It promotes critical and creative thinking skills, facilitates deep comprehension of texts, and enhances the ability to articulate ideas with clarity and precision, making it indispensable for academic achievement and professional success (Shaaban, 2006, p. 115).

4. Objectives of Linguistic Intelligence

Cultivate reading comprehension and linguistic analysis.

Enhance creative speaking and writing skills.

Foster critical thinking and clear expression of opinions.

Enable learners to communicate effectively in diverse real-life contexts (Mar'i & Al-Heila, 2010, p. 97).

5. Characteristics of Linguistic Intelligence

- A propensity for reading, writing, and a love of stories and literary texts

- Ability to manipulate words and employ them with precision

- High proficiency in persuasion and oral communication

- Facility in learning foreign languages

- Extensive use of vocabulary and linguistic symbols in expression (Zaitouni, 2014, p. 88)

Axis III: Relationship between Cooperative Learning and Linguistic Intelligence

Cooperative learning is one of the most effective instructional approaches for nurturing linguistic intelligence, as it relies on continuous verbal interaction among learners while working in groups. This interaction creates a rich environment for discussion, opinion exchange, questioning, and idea reformulation, all of which enhance learners' expressive language skills and reading comprehension (Johnson & Johnson, 1999, p. 45).

Cooperative learning also supports the development of writing and expressive skills by assigning groups tasks that require drafting reports or preparing written presentations, thereby providing broader opportunities for practical language use. Consequently, linguistic intelligence becomes a pivotal element in the success of cooperative learning, since the level of language proficiency directly influences the quality of communication and interaction within the group (Slavin, 2005, p. 132).

From an educational psychology perspective, the relationship between cooperative learning and linguistic intelligence is reciprocal: cooperative learning hones learners' language skills through dialogue and social interaction, while linguistic intelligence enhances the quality of cooperative learning by facilitating greater understanding and communication among group members (Mar'i & Al-Heila, 2010, p. 104).

Table1: Relationship between Cooperative Learning and Linguistic Intelligence

Aspect of Relationship	Manifestation in Cooperative Learning	Impact on Linguistic Intelligence
Verbal Interaction	Group discussions and questioning	Enhances oral dialogue and communication skills (Johnson & Johnson, 1999, p.45)
Problem-Solving Collaboration	Team-based academic task completion	Improves clarity and precision in idea expression (Slavin, 2005, p.132)
Report and Presentation Preparation	Joint written composition by group members	Boosts creative writing and linguistic thinking skills (Mar'i & Al-Heila, 2010, p.104)
Interdependence	Linking individual success to group success	Strengthens listening skills and reading comprehension (Zaitouni, 2014, p.88)
Social Interaction	Continuous peer communication	Increases vocabulary breadth and lexical richness (Shaaban, 2006, p.115)

Axis II: Practical Dimension—Application of Linguistic Intelligence and Cooperative Learning

This axis examines the practical relationship between linguistic intelligence and cooperative learning by deploying field research instruments and analyzing data collected from the study sample. It outlines the research methodology, data-collection tools, implementation procedures, and statistical methods used for result analysis, and explains how both constructs were measured in an authentic educational setting.

Research Methodology

The researcher adopted a descriptive-analytical design, appropriate for investigating the relationship between cooperative learning strategies and linguistic intelligence in second-grade students.

Population

The study population comprised all primary school teachers in the Directorate of Education of Djelfa Province during the 2024/2025 academic year.

Sample

A sample of 120 primary school teachers was selected from 15 schools across different regions of Djelfa Province. Of these, 114 completed the electronic questionnaire and constituted the study sample.

Study Instrument

An electronic questionnaire with 20 validated and reliable items (Cronbach's $\alpha = 0.695$) was administered online to the sample, yielding 114 valid responses for analysis.

4.1 Sample Characteristics

The gender distribution of participants was nearly balanced, with 52.6% female teachers and 47.4% male teachers, reflecting relative gender parity that may contribute to varied cooperative-learning practices and, in turn, positively influence students' linguistic intelligence.

Table 2. Educational Level Distribution among Second-Grade Teachers (N = 114)

Educational Level	Frequency	Percentage (%)
Bachelor's	21	18.4
Master's	39	34.2
Doctorate	54	47.4
Total	114	100

As shown in Table 2, the majority of participants hold a doctorate (47.4%), followed by master's degree holders (34.2%) and bachelor's degree holders (18.4%), indicating a predominantly advanced academic background that is likely to enhance the effective implementation of cooperative learning strategies and thus support the development of linguistic intelligence among students.

1.5 Instrument Validity

To verify alignment with the study's objectives and ensure the questionnaire's validity, the researcher consulted previous literature and calculated Cronbach's alpha. Table 1 presents these results:

Table 3. Overall Reliability of the Questionnaire (Cronbach's Alpha)

Value	Item
20	Number of items
0.695	Cronbach's alpha coefficient

As shown, the internal consistency of the questionnaire is acceptable ($\alpha = 0.695$), indicating that the instrument is reliable for the current study.

Instrument Reliability

SPSS 23 (Statistical Package for the Social Sciences) was used to compute frequencies, percentages, means, standard deviations, and t-tests.

1.7 Presentation and Interpretation of Results

Hypothesis 1: There are statistically significant differences in primary teachers' use of cooperative learning strategies by gender.

To test this hypothesis, means and standard deviations for male and female teachers' responses regarding strategy use were calculated, and an independent-samples t-test was conducted:

Variable	Gender	N	M	SD	t	df	Sig. (2-tailed)
Use of cooperative learning strategies	Male	54	36.56	5.04	3.514	112	0.001
	Female	60	33.30	4.07			
Impact of strategies on linguistic intelligence	Male	54	37.78	6.39	4.521	112	0.000
	Female	60	32.50	6.07			

Interpretation:

The t-test revealed significant gender differences in the reported use of cooperative learning strategies: males ($M = 36.56$) versus females ($M = 33.30$), $t(112) = 3.514$, $p = 0.001$, favoring male teachers. Similarly, males reported a greater perceived impact on linguistic intelligence ($M = 37.78$) compared to females ($M = 32.50$), $t(112) = 4.521$, $p < 0.001$. These differences may stem from variations in teaching style, confidence in managing group work, or access to training opportunities.

Presentation and Interpretation of Hypothesis 2

Hypothesis 2: There is a statistically significant positive correlation between the level of cooperative learning strategy use and its impact on developing linguistic intelligence, from teachers' perspectives.

Table 4. Descriptive Statistics for Strategy Use and Impact on Linguistic Intelligence (N = 114)

Measure	Mean	SD
Use of cooperative learning strategies	34.84	5.18
Perceived impact on linguistic intelligence	35.00	6.74

These close means suggest a positive association between teachers' strategy use and their observation of its benefits for students' linguistic intelligence. A subsequent Pearson correlation analysis will confirm this relationship statistically; a significant positive coefficient would support the hypothesis that increased cooperative learning correlates with enhanced linguistic intelligence.

Conclusion

This field study on primary teachers' perspectives confirms that cooperative learning is an effective approach for developing students' linguistic intelligence. When implemented with varied, well-planned activities, it enhances both oral and written communication skills and fosters critical and creative thinking through collaborative environments.

Gender differences emerged, indicating that male teachers reported higher usage and perceived impact of cooperative strategies compared to female teachers, possibly reflecting differences in professional training or pedagogical preferences. Moreover, the anticipated positive correlation between strategy use and linguistic intelligence highlights the importance of systematically integrating cooperative learning into educational practice.

These findings underscore the need to shift from traditional teaching methods toward learner-centered models that position students as active participants. Continuous professional development is essential to equip teachers with the skills required for effective cooperative learning implementation, with attention to individual learner differences and structured group dynamics to ensure inclusive participation and achievement of learning objectives.

Recommendations

Provide professional development for teachers on diversifying and adapting cooperative learning strategies to align with students' varying proficiency levels and abilities.

Integrate cooperative learning into teacher training curricula, including case studies and successful practical examples from real school settings.

Cultivate a supportive classroom environment that fosters dialogue and positive peer interaction.

Develop evaluation instruments that directly measure the impact of cooperative learning strategies on students' linguistic intelligence.

Involve students in collaborative language activities both inside and outside the classroom to strengthen their expressive and communicative skills.

Conduct longitudinal and cross-level studies to assess the broader effects of cooperative learning across different educational stages.

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